European Elites: An Exploration of LLM Methodologies for Surveying Elite Networks

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Abstract

We present a methodology for LLM-assisted research that anchors every claim to a specific sentence in a source document. The paper situates this methodological intervention within the diffuse field of elite research and the contested question of where power lies in the European Union. Using European institutional elites as a case study—commissioners, DGs, and MEP leaders across seven commission terms (1995–2029)—we demonstrate a pipeline that acquires and hashes controlled sources, segments them into sentences, uses LLMs to propose phrase-level claims, locates every quote deterministically, and submits uncertain evidence to a second adjudicator. The current build contains 5,490 claims, of which 4,144 are verified; 192 remain under review and 1,154 were rejected. The verified subset supports descriptive findings: commissioners concentrate in a small set of transnational policy organisations (ECFR, Friends of Europe, Bilderberg, Trilateral Commission), MEPs in parliamentary and party structures, and DGs in the College of Europe; education is organised around parallel institutional tracks rather than a single pipeline. We report on failure modes encountered—LLM hallucination, non-deterministic extraction, entity duplication, and the limits of string-similarity matching—and describe the methodological choices that address each.

1 Introduction

1.1 Elites in social theory

One of the signature features of the research agenda driven by structural-demographic theory has been its emphasis on elites, and how they figure in the dynamics leading to societal destabilisation, crisis and sometimes collapse. Needless to say, SDT researchers have not been the first to single out elites for social scientific attention. Indeed, there is little doubt that in one way or another, elites have represented an area of considerable interest for students of society and politics since the very dawn of this domain of human inquiry, as attested by Plato's argument for government by philosopher-kings and Aristotle's detailed commentaries on the relative merits and demerits of oligarchy, aristocracy and democracy.

However, in the era of modern social science, elites have been the subject of more dedicated and sustained attention since at least the time of the Italian elite theorists of the late nineteenth and early twentieth centuries—authors from whom we have inherited concepts such as the "organised minority" (Mosca), the "circulation of elites" (Pareto), and the "iron law of oligarchy" (Michels) [2].

Subsequent generations of social scientists have looked at elites from somewhat different angles again. From the mid-twentieth century onwards, studies such as those of C. Wright

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Mills and G. William Domhoff asked searching questions about who truly governed, and about where the agency behind crucial decisions truly lay in polities where elected officials co-existed with powerful unelected (and often non-public-facing) elites. These arrangements were conceptualised in different ways: Mills emphasised the overlap between political, military and corporate elements in the "power elite" that shaped US governance [7]; Domhoff promoted a class-domination theory of power; Michael J. Glennon described the bifurcation between the outward-facing "Madisonian" institutions and the more obscure "Trumanite" network in terms of "double government"; while Peter Dale Scott spoke of "deep politics" or simply the "deep state". However, what these perspectives had in common was the idea that the reality of power looked very different from the pluralist models associated with leading political scientists such as Robert Dahl.

Meanwhile on a more sociological level, scholars such as Pierre Bourdieu and his many acolytes have focused on the processes (often mediated through elaborate cultural and symbolic codes) through which elites ensure social reproduction across generations.

1.2 Elites and the European project

More recently, political scientists working on both sides of the Atlantic have charted various routes through which political elites have seemed to increasingly insulate themselves from the cut and thrust of what used to be considered the basic operations of everyday democratic politics, leaving "cartel parties" and a hollowed out political "void" in their wake (Mair). Some of this research has been focused on the national level, while some of it has honed in on the role of supranational institutions as key sites in the distancing of political power from democratic checks and balances as traditionally understood. Some of it has charted the emergence of a new supranational European elite on the back of the institution-building processes, as Verzichelli and colleagues document [1]. Perhaps the best-documented example of this purported phenomenon is that of the European Union and its component organs, which has spawned an extensive scholarly literature and become something of a byword for democratic deficit—at least in the eyes of its critics, and sometimes, somewhat grudgingly, even in the eyes of its supporters. Much of this EU-oriented literature has been focused on the workings of the European institutions—sometimes casting them explicitly as venues where elite-level decision-making can be hived off from the democratic mechanisms notionally built into national domestic politics.

This has dovetailed with recent research by a number of political scientists based in the US who have deployed fine-grained quantitative research tools to interrogate that polity's purported democratic credentials, only to find that US government policy appears to be shaped overwhelmingly by elite preferences, while the preferences of non-elites seem to exert negligible influence (Gilens and Page). Though not explicitly framed by the researchers in terms of "elite theory", their work is clearly pertinent in this regard.

Comparative work on European power elites likewise finds pronounced social immobility [2].

Taken together, all of this might seem to indicate the existence of a thriving ecosystem of elite research, whatever its implications for the socio-political health of contemporary societies. However, the relative abundance of research pertaining in one way or another to elites belies a remarkably diffuse field of enquiry. Across this landscape of elite-focused studies, we find research questions that are not just distinctly different from one another, but sometimes in certain important respects pulling in opposite directions. Some of these questions concern the perpetuation of elite dominance. Others, arguably, are about something close to its inverse. Even within the narrower field of specifically SDT-driven research, we find distinctly different kinds of questions about elites (and would-be elites) being formulated. Some of these are concerned with what elites try to do (often unsuccessfully) to stave off crisis and collapse in tumultuous times. Others are concerned with how the overproduction of elites or elite aspirants directly *contributes* to societal destabilisation. This latter seam of research can be split, in turn, into at least two rather distinct substrata. To put it somewhat crudely, this means looking not

just at elites as "winners", but at elites (or perhaps more accurately "would-be", "wannabe" or "has-been" elites) as "losers". Both phenomena can be highly destabilising. As Turchin has pointed out, the overproduction of the elites that we might call "winners" tends to result in the ratcheting up of the "wealth pump", leading to unsustainable levels of extraction from the wider economy and society, and the gradual impoverishment of non-elites. Meanwhile, the overproduction of "elites as losers" poses its own problems, which can take a number of forms. One example might be that of the *ancien régime* French nobles who lashed out against the crown's attempted reforms in the run-up to the crisis of 1789, seeing them as only being likely to speed up the trajectory of downward social mobility they perceived themselves to be on, thereby unwittingly undermining the remaining supports of the system that they clung to, limpet-like, for sustenance, and ultimately accelerating France's slide into revolution and civil war. Another example might be the heavily indebted graduates of US law schools and other advanced degree programmes who are currently trying and failing to turn their credentials into lucrative careers, and increasingly breaking away from the traditional political formations to support more radical offerings—identified by Turchin and other SDT researchers as a recipe for serious instability.

Thus "elite research" is not one thing, but many things, and any examination of this shorthand immediately throws up fundamental questions, including but not limited to the following: Whom are we really talking about when we talk about "elites"? How many different types of "elite" are there, and who should really be called "elite"? In so far as "elite" is a meaningful concept, how does one attain this status, or maintain it in an era in which many individuals with "elite" status are experiencing downward social mobility and becoming *déclassés*? Relatedly, how does one lose elite status? Do traditional means of attaining or maintaining elite status, such as acquisition of higher education (cf the famous "college premium"), still apply, or is this now being washed away in a sea of credentialism, degree inflation and unsustainable student debt? What is the relation between the "winner" and "loser" elites? Is there a firm distinction between the two, or more of a revolving door?

And perhaps most importantly: how do these various questions relate to the moment of profound and multifaceted crisis that Western societies are currently experiencing?

1.3 Why the European Commission?

To be sure, this represents a tangled skein of questions rather than a strict research agenda, and while we cannot address all of them in our paper, we do propose to at least bring together a number of distinct strands of elite research as they pertain to Europe's current conjuncture. In so doing, we hope to achieve a number of goals. As well as bridging the gap between these somewhat disparate strands of elite theory, we aim to apply the resultant framework to relatively novel subject matter. Previously, SDT-inspired elite research has been applied to various historical and contemporary societies, but never, to our knowledge, to a supranational entity or supranational elite. Something similar could be said of "power elite" research along the lines of Mills or Domhoff. No consensus may exist as to how to explain what happens in the inner sanctums of American power, but at least we have some reasonably firm hypotheses to choose from, and to build upon. How power works at the level of Europe, meanwhile, is notoriously something of a black box.

This is not to draw an equivalence between American and European power. While the EU has long been recognised as a behemoth in areas such as trade, scholars have continuously debated the extent to which it can be considered a coherent political actor in a larger sense. Indeed, some scholars argue that, insofar as the EU is a power on the international stage, it is a declining one. Nonetheless, declining or otherwise, one cannot attempt to make sense of the current conjuncture without attempting to make sense of the EU and the elites that animate it. Whatever one thinks about the efficacy of the EU, it is no longer possible to think about German or French policy, either on a domestic or foreign level, in isolation from European

policy. Nor is it possible to think through the reactivation of Europe as a major theatre in the current not-so-cold war between East and West without considering the EU's role.

Of course, part of the reason that the inner workings of the EU are so obscure is that the institutional architecture of the European project is unusually complex. "Studying the EU" could mean many different things, and the first questions one faces when attempting to do so are "how?" and "where to begin?". Among the various institutions that comprise the EU, we have chosen to focus our initial analysis on the European Commission, as we believe that if any one entity embodies the supranational elite that is emerging on the back of the European project, it is this more or less permanent central bureaucracy. (Not for nothing is the Commission sometimes described as the beating heart of the EU.) However, we acknowledge that this basic starting premise of our analysis is open to a certain degree of debate, insofar as no consensus exists in the scholarly literature as to where power lies within the European project, or even as to the European project's basic nature. These questions have been the subject of many decades of intense debate, and while we cannot make them the focus of this paper, we do need to briefly address their basic outlines, insofar as they pertain to the current analysis.

Put simply, the various schools of thought on the nature of the European project, while manifold, can roughly be divided into two broad camps—that which understands the EU as a fundamentally inter-governmental initiative, and that which views it as an experiment in supranational governance. Furthermore, this difference in perspective is not merely scholarly or theoretical: the EU clearly incorporates both inter-governmental and supranational elements and, what is more, is riven by tensions over this very issue. That is, the theoretical difference in opinion maps onto a real-life one, in that the protagonists of the European project have themselves chronically disagreed over whether it ought primarily to be an intergovernmental or supranational enterprise. And while the actually existing structures incorporate both logics, it is difficult to say conclusively which is more important, as those structures have evolved on the back of multiple compromises, via multiple rounds of treaty negotiations, between actors with rivalling visions as to which logic ought to prevail. Thus this tension is woven into the basic fabric of the European institutions, and snapshots taken at different historical junctures since the founding of the European institutions would show the battle lines drawn in different ways between the intergovernmentalists and the supranationalists, with the initiative periodically slipping back and forth between the two dispositions.

In short, the question of where power lies in Europe is shrouded in some opacity, and we do not need to reach for the scholarly literature to see evidence of this. It is reflected in many of the stock phrases customarily used in real life to describe the workings of the European project. Indeed, when facing criticisms from "Eurosceptic" political actors who lament the eclipse of national sovereignty, their "pro-European" counterparts cheerfully acknowledge that the EU is built on a principle of "pooled sovereignty". But it is not always clear where exactly this sovereignty "pools together", as it were. Indeed, an oft-repeated line widely attributed to Henry Kissinger reflects this opacity. In reference to the diffuse nature of the European power structure, Kissinger is purported to have quipped: "Who do I call when I want to speak to Europe?"

On the other hand, some commentators would suggest that, if US leaders ever really did struggle to "speak to Europe", as Kissinger is said to have suggested, this issue would seem to have been less salient in recent times. From this point of view, Europe has, in spite of increased trans-Atlantic verbal sparring in the Trump era, largely abandoned the independentist streak of the 1990s and early 2000s and increasingly fallen into line with the United States, at least on key foreign policy issues.

To the extent that this is true—and clearly the point is controversial—it raises a further difficult question for anyone attempting to carry out research on the EU elite, and another dimension to the question of where power lies. That is, we are no longer merely pondering whether power really lies in the Commission or in the Council, or in Brussels, Frankfurt, or the

various national capitals. We also must take seriously the question of whether it even lies in Europe.

In sum, the study of "elites" is the study of many different things: rising elites, declining elites, would-be elites, entrenched elites, "power elites" and "elites" who may not yet be obviously experiencing downward social mobility, but whose real power is open to question. In turning our focus to the cadres of the European Commission, we leave open the question of exactly what kind of elite it is. However, taking it that it is an elite of some kind, and that shedding light on how it is constituted would be useful in the current conjuncture, we take a preliminary step toward this goal in the following sections.

1.4 From elite theory to a citation-anchored method

The step we take is methodological. The evidentiary challenge follows from the conceptual one. Elite formation is distributed across 27 member states with distinct educational systems, career pathways, and political cultures; the networks that connect these elites cross national boundaries and institutional sectors [3]; and the relevant evidence is scattered across biographies, CVs, declarations of interest, and institutional records. Capturing it requires aggregating data across languages, jurisdictions, and source types, while keeping every claim traceable to its source.

The remainder of this paper describes a methodology that treats the LLM as an extraction engine, not an authority. Every claim in our database is linked to a specific sentence in a specific document. The LLM's output is a hypothesis, verified against the source before acceptance. The current build contains 5,490 claims, of which 4,144 are verified, across 3,647 entities and 820 hashed source documents. It supports analysis of seven European Commissions from Santer (1995) through Von der Leyen II (2024–2029).

2 Methodology

LLMs hallucinate because they generate text by sampling from a learned probability distribution over tokens. When the model lacks grounding in external facts, this distribution can produce plausible-sounding but incorrect statements [4]. Manakul et al. (2023) demonstrate that hallucinated outputs are detectable through self-consistency: when an LLM is asked the same question multiple times, hallucinated answers tend to diverge and contradict each other, while factual answers remain consistent across samples [6]. This reveals that hallucination is a property of *unconstrained* generation: the model's internal distribution over tokens is underdetermined by the prompt, and stochastic variance produces divergent outputs.

Crucially, Kadavath et al. (2022) show that LLMs are substantially better at *evaluating* their own outputs than at *generating* correct ones—they can detect errors in existing text at much higher rates than they can avoid producing errors during generation [5]. This generation-verification gap is the foundation of our approach: we do not ask the LLM to invent facts. We ask it to cite a numbered phrase from a source document that supports a factual claim. The task changes from "produce text from your internal distribution" to "does this specific sentence support this specific claim?"—a form of natural language inference with dramatically lower hallucination rates.

LongCite (2024) provides direct evidence that LLMs can reliably anchor claims to specific source sentences when trained and prompted to do so [8]. Our methodology extends this insight: by numbering phrases and requiring the LLM to return a phrase index for each extracted fact, we build a system in which every claim in the database is traceable to a specific sentence in a source document. The LLM acts not as a knowledge base but as a structured extraction engine with mandatory provenance.

2.1 System Architecture

The pipeline has six stages: acquire and hash sources; segment each source into sentences with exact character offsets; propose candidate claims; locate and grade their evidence; adjudicate uncertain evidence; and resolve entities and render the graph. Each stage writes a durable artifact: a source snapshot, a sentence table, claim and evidence rows, assertion decisions, and a graph export. The first two stages replace the v1 assumption that a phrase index could be reused across tokenisers.

2.2 Database Schema

The database is a provenance graph on a relational backend. Its core tables are:

- `source_document`: one snapshot per source, with URL, local path, SHA-256, revision, retrieval date, and access date.
- `sentence`: exact character offsets; sentence IDs, not list positions, are the only locator used downstream.
- `entity` and `entity_alias`: persons, organisations, institutions, commissions, and countries, with aliases for resolution.
- `claim`: a candidate triple—subject, predicate, object, and role. A claim is not yet a finding.
- `evidence`: the claim, source document, sentence span, exact quote, locator status, and flags for subject, object, and role.
- `assertion`: the decision—verified, needs_review, or rejected—with an evidence grade of A, B, C, or none. Only verified assertions enter the graph.
- `adjudication`, `rejection`, and `review`: recorded model verdicts, rejection reasons, and human decisions.
- `organisation_profile`: sourced or manual-seed organisation attributes, kept separate from graph edges.

A verified assertion must have at least one located evidence row. A claim whose quote cannot be found in the cited document is rejected rather than assigned a guessed index. The v2 invariant checker (`check_v2.py`) enforces this, together with foreign keys, source hashes, and the absence of duplicate active entities. This replaces v1's `fact`/`provenance` pair and its claim that facts without provenance were simply deleted.

2.3 Phrase-Level Extraction

Documents are segmented once into sentences with exact character offsets. The corpus is not truncated: long biographies are chunked, and every sentence keeps its ID. The LLM receives a numbered chunk and returns an organisation, a role phrase, a relation, the exact evidence sentence, and the sentence number. The evidence string is then located deterministically in the source; if it is not found, the claim is rejected. A separate grading step checks whether the located span contains the subject, object, and role; grade A is deterministic, grade B has an implicit role, and grade C is sent to adjudication. This replaces the v1 practice of storing a phrase index produced by a different tokeniser from the one that generated the evidence.

2.4 Official Source Acquisition

Wikipedia biographies supplied broad initial coverage, but a substantial minority of people had no local source at all. We therefore acquired official institutional documents directly: 26 member biographies from the Court of Justice of the European Union (CURIA), six Commission Director-General and Deputy Director-General CVs or person pages, and five corporate leadership biographies from Eni, SAP, Safran, and Neoenergia. Each document was fetched from its institutional URL, converted to text, hashed, segmented, and attached to its subject; extraction then ran through the same claim, evidence, and assertion path as every other source. These acquisitions raised verified education coverage from 74/85 to 78/85 for DGs and DDGs, from 59/79 to 64/79 for corporate elites, and from 18/37 to 22/37 for CJEU members.

Fifteen people remain without a source: seven CJEU judges, one advocate general, four MEPs, and three corporate elites. In most cases the institution publishes no CV or biography for the person, or the page is unavailable to automated retrieval; one corporate case is a duplicate entity under two name forms. These are recorded as missing evidence, not as negative findings.

2.5 Failure Modes

2.5.1 The Short-Extract Problem

Our initial v1 extraction runs used Wikipedia API extracts of 300–400 characters. The LLM hallucinated freely: it returned the commissioner's own name as an educational institution and produced generic names such as "aeronautical engineer". Re-scraping the same biographies as full texts produced clean extractions with specific institutions. The lesson is that the input, not the model's capability, was the immediate cause. The v2 corpus therefore stores full texts and sentence offsets, and rejects claims whose quotes cannot be located.

2.5.2 Non-Deterministic Extraction

Identical prompts can produce different candidate sets. V2 treats extraction output as a candidate pool rather than a finding: manifests and caches are versioned, each run records its model and parameters, and the assertion layer is rebuilt from the located evidence. This makes a rerun auditable even when the candidate set changes.

2.6 Entity Resolution and the Limits of String Similarity

Extraction produces organisation names as they appear in source texts: "Radio Netherlands", "Radio Netherlands Worldwide", "FEPS", "Foundation for European Progressive Studies". String similarity alone is dangerous: "CFR" and "ECFR" score 0.857 on difflib, yet one is the US Council on Foreign Relations and the other the European Council on Foreign Relations. Pure auto-merge below 0.95 would collapse them.

V2 resolves names with a deterministic merge key that strips parenthetical qualifiers, normalises punctuation and accents, and preserves sub-entities such as an "EPP group" as distinct from the party. The registry carries aliases, stable slugs, external identifiers where available, and a quarantine status for junk surface forms. Residual fuzzy variants are listed for review rather than silently merged. The earlier v1 LLM judge is no longer the merge authority; it remains useful only as an advisory signal. Every final merge is reproducible from the registry.

2.7 Organisation Profiles

Organisation type, funding, headquarters, and descriptions are stored in `organisation_profile`, separate from graph edges. The current seed is manually researched and marked `manual_seed`; it is not presented as a sourced assertion. The v2 graph does not use LLM-generated classifications for colouring or analysis. Entity categories come from the registry, and uncertain classifications

remain in the review queue. This replaces the v1 practice of treating LLM classifications as evidence.

3 Wiki Output and Human Review

The verified layer renders to a static wiki with 3,320 pages: 459 people, 2,355 organisations, 7 commissions, 495 source documents, plus index, statistics, and citation pages. Each page shows only verified assertions, with role, date, quote, evidence grade, and a link to the source document. Internal links are Obsidian wikilinks; source pages are filtered from the default graph view, so the browsing graph is not dominated by document nodes.

The typed graph is exported separately: 2,960 nodes and 3,702 edges, of which 2,613 are network edges, 632 education, 361 occupational, 94 institutional, and 2 other. The graph is not an Obsidian screenshot: `nodes.csv`, `edges.csv`, and `graph.graphml` carry edge class, predicate, role, grade, quote, document path, SHA-256, and a URL where the source is a public web page. Obsidian is a browsing view; the analysis uses the export.

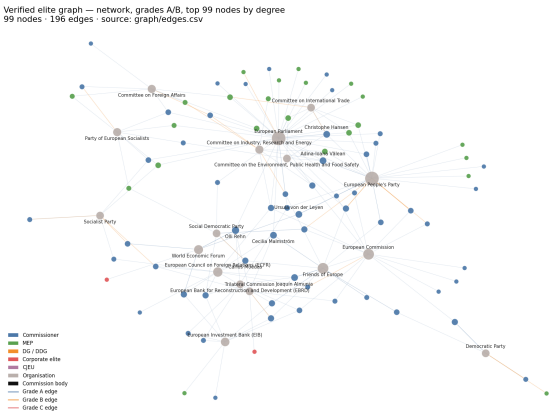


Figure 1: Verified elite network (network edges, grades A–B; top nodes by degree). Commissioners are blue, MEPs green, DGs orange, corporate elites red, CJEU members purple, and organisations grey.

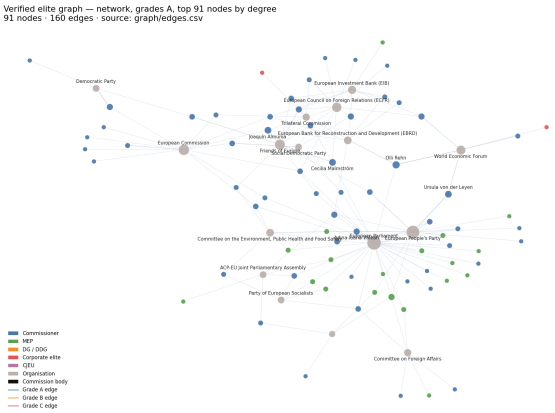


Figure 2: The same export restricted to grade-A edges. The stricter view is used as a sensitivity check on the network findings.

The review layer is explicit. Assertions that the deterministic checks cannot settle are marked `needs_review`; grade-C claims are rechecked by a second judge, and disagreements are not silently promoted. In the current build, 569 grade-C claims were kept by two-model

consensus and 169 were moved to review. The queue is exported as a priority-scored table, and every human decision is recorded with reviewer, reason, and timestamp.

4 Findings

The following findings describe the verified subset, not the larger candidate pool. The dataset covers 150 commissioners, 162 MEP-only entities (168 MEPs in the frame; some are merged with commissioner records), 85 DGs and DDGs, 79 corporate elites, and 37 CJEU members. Coverage is uneven: commissioner education is verified for 119/150, DG/DDG for 78/85, MEP for 91/162, corporate for 64/79, and CJEU for 22/37. We therefore report descriptive patterns and avoid claims that depend on the unverified remainder.

4.1 Elite Network Structure

Commissioners share a policy-institute cluster: the European Council on Foreign Relations (12 commissioners), Friends of Europe (10), Bilderberg (9), and the Trilateral Commission (7). MEPs sit in parliamentary and party structures: the European Parliament (23), the European People's Party (11), the Party of European Socialists (8), with committee memberships at 5–6 each. DGs and DDGs centre on the College of Europe (8). Corporate elites are more fragmented: the largest shared organisation is the European Round Table of Industrialists (3), and 8 of 79 share an elite-network organisation with a commissioner under our broad list.

Both groups connect to parties and the Parliament; the commissioner cluster is distinguished by the policy institutes, not by an exclusive network.

4.2 Education Tracks

Table 1 shows the eight institutions with six or more verified attendees, broken down by elite track.

Institution	Political	Admin.	Judicial	Corporate	Total
College of Europe	3	8	1	0	12
Harvard	6	2	1	0	9
École Polytechnique	0	0	0	7	7
University of Cambridge	2	3	1	0	6
London School of Economics	4	1	0	1	6
Sapienza University of Rome	4	0	0	2	6
Sciences Po	6	0	0	0	6
University of Oxford	3	1	0	2	6

Table 1: Elite school specialisation by verified track. Political = commissioners + MEPs; Admin. = Directors-General and deputies; Judicial = CJEU judges and advocates general; Corporate = transnational corporate leaders.

The College of Europe is the most balanced pipeline, spanning political, administrative, and judicial tracks with an administrative weight. Harvard and Cambridge are politically weighted but also span administrative and judicial tracks. Oxford, the London School of Economics, and Sapienza combine political attendance with either administrative or corporate tracks. Sciences Po is exclusively political in the verified record, and École Polytechnique is exclusively corporate. The data support a set of parallel institutional tracks rather than a single elite pipeline.

4.3 The French Pipeline in the Verified Record

Among French commissioners with verified education, Pascal Lamy (ENA, HEC, Sciences Po) and Pierre Moscovici (ENA, Sciences Po) follow the Parisian *grandes écoles* route, while Stéphane Séjourné's record contains only the University of Poitiers. Jacques Barrot has no verified education in the current build. The verified record therefore preserves the contrast, but it is too sparse to support a commission-by-commission decline; we do not report such a figure.

5 Future Work

Coverage remains uneven: commissioner education 119/150 and organisation affiliation 137/150; DG/DDG 78/85 and 20/85; MEP 91/162 and 118/162; corporate 64/79 and 29/79; CJEU 22/37 and 2/37. Three tasks follow. First, work the 192 `needs_review` assertions and run the grade-A second judge. Second, acquire the missing sources for the corporate, CJEU, DG/DDG, and MEP populations rather than treating their absence as negative evidence. Third, add a formal overrepresentation metric based on Eurostat tertiary education statistics, so claims of elite signalling can be tested against population baselines. Extending the frame to General Court judges and national officials is a further source-acquisition task.

6 Conclusion

This paper's own bibliography was verified using the methodology it describes. Each cited work was downloaded as a PDF, checked by `check_citations.py`—a script that extracts citation claims from the LaTeX source, searches the cited PDF text, and sends numbered phrases to an LLM for phrase-level verification with confidence scoring. At the time of writing, all citations pass the checker's verification threshold. The verification report is committed alongside this paper and can be regenerated by any reader with API access.

The key methodological claim of this paper is that LLM-assisted research can be rigorous if the LLM is treated as a hypothesis generator rather than an authority. The specific mechanisms—hashed source snapshots, sentence-level locators, deterministic quote resolution, evidence grades, recorded adjudication, and a human review queue—are transferable to other domains. The case study demonstrates that this discipline yields auditable data and descriptive findings, while also showing which earlier claims do not survive stricter verification. That is the appropriate outcome for a methodology paper: the evidence layer is designed to make such revisions visible.

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